VF Asia Footwear Laboratory Testing Manual

Test Method for Non- Volatile Content	Test method # VF-A08 Page 1 of 2
Date Issued : May 27, 2008 Revision # :	Created by : Nila Babia Erazo /

PURPOSE AND SCOPE: To measure the percentage of non- volatile ingredients contained in a solvent or water-based adhesives, primers, and ink.

REFERENCES: ASTM D2639

EQUIPMENT AND MATERIAL:

1. Balance, capable of weighing to 0.0001grams.
2. Weighing dish, aluminum or plastics.
3. Air-circulating oven, capable of maintaining 110.0 ± 2.0 ° C.
4. Desiccator, with drying agent.
5. Solvent, water – based adhesives, primers, coatings, etc.

SAMPLE PREPARATION:

Test should be carried out at the standard laboratory condition at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

Ensure that a well-mixed, homogeneous test material is used. Effective mixing can be performed by manual stirring or suitable mixing equipment.

REPETITIONS: Test two (2) samples per material

TEST PROCEDURE:

Solvents used for adhesive, primers and inks are flammable which should be handled carefully. Contact with the skin and inhalation must be avoided. Protective gloves, gas mask and safety glasses should be worn.

1. Weigh the weighing dish to the nearest 0.0001 grams **(W1).**
2. Pour 0.5 ± 0.1 grams of a test material into the weighing dish.
3. Immediately obtain and record the weigh of a test sample and weighing dish to the nearest 0.0001 grams **(W2).**
4. Place the test sample/weighing dish in an oven at 110 ° C for 30 minutes.

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5. Remove the test sample /weighing dish from the oven. Place in a desiccator, and allow to cool to 23.0 ± 2.0 ° C.
6. Weigh the test sample / weighing dish to the nearest 0.0001 grams **(W3).**

CALCULATION:

Non- volatile content (%) = (W3-W1) / (W2-W1) x 100

REPORT: The average of the two readings to the nearest 0.1 percent (%)